

REQUIREMENT ANALYSIS AND CUSTOMER DATABASE MODULE

Requirement Analysis:

An e-commerce system manages the complete process of buying, selling, and managing products online. The system allows customers to search for products, add products to the cart, place orders, make payments, and receive deliveries.

Process Flow:

- Browse / Search Product – The customer searches for the product they want to purchase.
- Add to Cart – After selecting the required product, the customer adds it to the cart.
- Order Placed – The customer selects the delivery address and places the order.
- Checkout / Payment – The customer proceeds to the payment section and completes the checkout process.
- Delivery – After successful payment, the product is assigned for dispatch and delivered to the customer.
- Review & Support – After delivery, the customer can provide a review. They may also request support for installation, return, or replacement.

Entities:

Customer: The Customer is the end-user who browses and purchases products.

Key Attributes:

- customer_id (Primary Key)
- first_name
- last_name
- email
- phone
- created_at

Relationship: One-to-Many with Order and Review.

Seller: The Seller is a merchant or vendor who lists products on the platform.

Key Attribute

- seller_id (Primary Key)
- company_name
- email
- phone
- rating

bank_details:

Relationship: One-to-Many with Product.

Product: The Product is an item listed for sale on the platform.

Key Attributes:

- product_id (Primary Key)
- seller_id (Foreign Key)
- title
- description
- price
- stock_quantity
- SKU

Category:

Relationship: Many-to-One with Seller and One-to-Many with Review.

Order: The Order represents a transaction made by a customer.

Key Attributes:

- order_id (Primary Key)
- customer_id (Foreign Key)
- order_date
- status
- total_amount
- shipping_address

Relationship: Many-to-One with Customer, One-to-One with Payment, and Many-to-Many with Product through OrderItem.

Payment: The Payment represents the monetary transaction associated with an order.

Key Attributes:

- payment_id (Primary Key)
- order_id (Foreign Key)
- payment_method
- status
- transaction_id
- amount
- payment_date

Relationship: One-to-One with Order.

Review: The Review contains feedback and rating submitted by a customer for a product.

Key Attributes:

- review_id (Primary Key)
- customer_id (Foreign Key)
- product_id (Foreign Key)
- rating
- comment
- created_at

Relationship: Many-to-One with Customer and Product.

Customer Table Creation:

The Customer table is created to store customer details such as customer ID, customer name, email, phone number, address, and registration date.

Describe Customer Table:

The DESC CUSTOMER command is used to display the structure of the Customer table. It shows the column names, constraints, and data types.

```
SQL> CREATE TABLE CUSTOMER(customer_ID INT PRIMARY KEY, customer_Name VARCHAR(100) NOT NULL, Email VARCHAR(100) UNIQUE, Phone NUMBER(15) UNIQUE, Address
VARCHAR(200), Registration_Date DATE NOT NULL);

Table created.

SQL> desc customer
Name                               Null?    Type
-----
CUSTOMER_ID                        NOT NULL NUMBER(38)
CUSTOMER_NAME                      NOT NULL VARCHAR2(100)
EMAIL                             VARCHAR2(100)
PHONE                             NUMBER(15)
ADDRESS                           VARCHAR2(200)
REGISTRATION_DATE                  NOT NULL DATE
```

Insert Customer Values

```
SQL> INSERT INTO Customer VALUES (102, 'kavya', 'kavya@gmail.com', 9876543221, 'coimbatore', TO_DATE('2026-03-10', 'YYYY-MM-DD'));

1 row created.
```

```
SQL> INSERT INTO CUSTOMER VALUES('&Customer_ID', '&Customer_Name', '&Email', '&Phone', '&Address', TO_DATE('&Date', 'YYYY-MM-DD'));
Enter value for customer_id: 101
Enter value for customer_name: AJAY
Enter value for email: ajay@gmail.com
Enter value for phone: 9876543220
Enter value for address: chennai
Enter value for date: 2026-03-01
old 1: INSERT INTO CUSTOMER VALUES('&Customer_ID', '&Customer_Name', '&Email', '&Phone', '&Address', TO_DATE('&Date', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('101', 'AJAY', 'ajay@gmail.com', 9876543220, 'chennai', TO_DATE('2026-03-01', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 102
Enter value for customer_name: kavya
Enter value for email: kavya@gmail.com
Enter value for phone: 9876543221
Enter value for address: coimbatore
Enter value for date: 2026-03-10
old 1: INSERT INTO CUSTOMER VALUES('&Customer_ID', '&Customer_Name', '&Email', '&Phone', '&Address', TO_DATE('&Date', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('102', 'kavya', 'kavya@gmail.com', 9876543221, 'coimbatore', TO_DATE('2026-03-10', 'YYYY-MM-DD'))

1 row created.

SQL> /
Enter value for customer_id: 103
Enter value for customer_name: MANOJ
Enter value for email: manjo@gmail.com
Enter value for phone: 9876543222
Enter value for address: Trichy
Enter value for date: 2026-03-05
old 1: INSERT INTO CUSTOMER VALUES('&Customer_ID', '&Customer_Name', '&Email', '&Phone', '&Address', TO_DATE('&Date', 'YYYY-MM-DD'))
new 1: INSERT INTO CUSTOMER VALUES('103', 'MANOJ', 'manjo@gmail.com', 9876543222, 'Trichy', TO_DATE('2026-03-05', 'YYYY-MM-DD'))
```

Display Customer Records

```
SQL> SELECT * FROM CUSTOMER;
```

CUSTOMER_ID	CUSTOMER_NAM	EMAIL	PHONE	ADDRESS	REGISTRATI
101	AJAY	ajay@gmail.com	9876543220	chennai	01-MAR-26
102	kavya	kavya@gmail.com	9876543221	coimbatore	10-MAR-26
103	MANOJ	manjo@gmail.com	9876543222	Trichy	05-MAR-26
104	NISHA	nisha@gmail.com	9876543223	Erode	15-MAR-26
105	RAVI	ravi@gmail.com	9876543224	chennai	20-MAR-26

Update Customer Details

```
SQL> UPDATE CUSTOMER SET PHONE = 9835274301 WHERE CUSTOMER_ID= '104';

1 row updated.
```

This operation successfully changes the phone number of the specified customer.

Conclusion:

The Customer Database Module provides a structured way to store and manage customer information in an e-commerce system. The Customer table contains important details such as customer ID, name, email, phone number, address, and registration date. Using SQL commands such as CREATE, DESC, INSERT, SELECT, and UPDATE, customer records can be created, viewed, inserted, and modified efficiently.